

# Multi-Objective Optimization: my journey over the past 35 years

28th International Conference on Multiple Criteria Decision Making  
University of Wuppertal, Germany

---

Xavier GANDIBLEUX  
Nantes Université – France  
May 2026

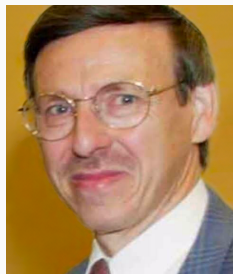
# Foreword

---

This research journey would not have been possible  
without the guidance, support, and inspiration  
of remarkable figures from the MCDM Community!

# Among my Professors at the University

Shaped my profile:



**Jean-Claude Vansnick**

Université de Mons, BE

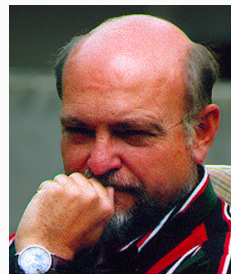
OR. Decision analysis



**Jacques Teghem**

Faculté Polytechnique de Mons, BE

MIP. Combinatorial optimisation



**Marc Roubens**

Faculté Polytechnique de Mons, BE

Statistics. Preference modelling

# During my PhD thesis

Hosted and advised me:



**Bernard Roy**

Université Paris-Dauphine, FR

Director of the LAMSADE



**Jean-Charles Pomerol**

Université Pierre et Marie Curie, FR

Director of the LIP6

# During my PhD thesis

Influenced my research trajectory:



**Pekka Korhonen**

Aalto University, FI

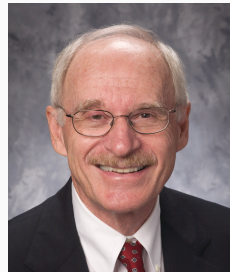
Interactive approaches



**Valerie Belton**

University of Strathclyde, UK

Decision support systems



**Ralph Steuer**

University of Georgia, USA

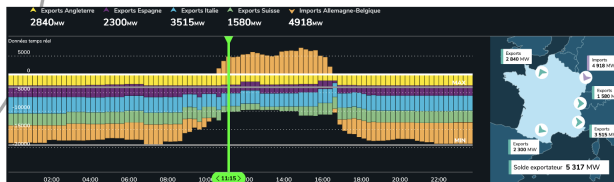
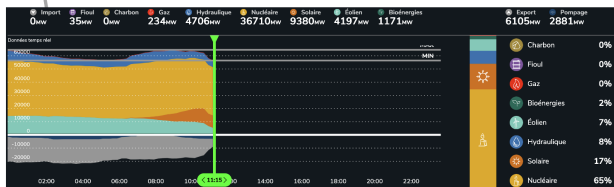
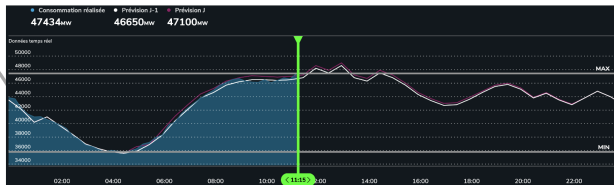
Multi-objective optimization

|                                                      |         |
|------------------------------------------------------|---------|
| Scalarizing Functions, Interactive Procedures, MCDSS | (1989–) |
| MOCO, MOMH, MCDMlib                                  | (1995–) |
| $X_E$ and $Y_N$ , Branch-and-Bound, Exact Algorithms | (2000–) |
| Open-Source Software                                 | (2014–) |
| Now and Tomorrow                                     | (2026–) |

Scalarizing Functions,  
Interactive Procedures, MCDSS  
(1989–)

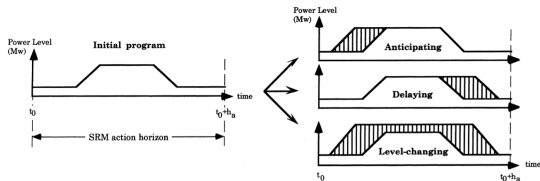
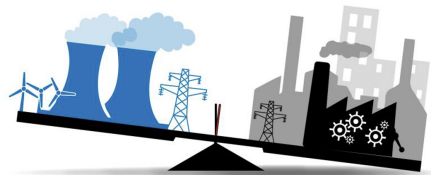
---

# Context: French national production of electricity



# Context: Disturbance and Need of Spinning Reserve Mobilization

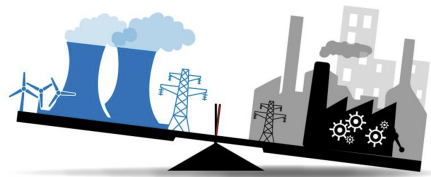
- ▶ Sudden disturbance creates imbalance between production and consumption
- ▶ When automatic reserves are insufficient
- ▶ Dispatcher must restore system via tertiary reserve (SRM)



- ▶ Short-term dispatch of production units
- ▶ Highly combinatorial decision space
- ▶ Time-critical: decisions must be made in minutes

# Context: Disturbance and Need of Spinning Reserve Mobilization

- ▶ Sudden disturbance creates imbalance between production and consumption
- ▶ When automatic reserves are insufficient
- ▶ Dispatcher must restore system via tertiary reserve (SRM)

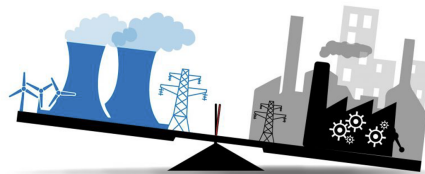


If it fails → **blackout !**

- **Spain**, April 2025 (15 gigawatts lost in five seconds): Spain, Portugal, southern France, until 1 day.
- **Germany**, November 2006: 10 million people, including in France, Germany, Belgium, the Netherlands, Italy and Spain, for nearly 1 hour
- **USA** (Ohio), August 2003: 50 million people between Toronto and New York, until 4 days
- **Italy**, September 2003: 56 million people in Italy, until 2 days
- **France**, December 1999: 3.6 million French households

# Solution: Multi-criteria decision support

- ▶ Sudden disturbance creates imbalance between production and consumption
- ▶ When automatic reserves are insufficient
- ▶ Dispatcher must restore system via tertiary reserve (SRM)



- ▶ Alternatives: possible actions on production units
- ▶ Constraints: elapsed time, generation limits, ramp rates, reserve constraints,...
- ▶ Criteria: cost, contribution, destorage of reserve, reliability...

## Trade-offs:

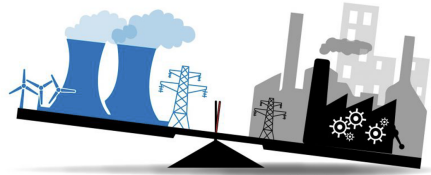
- rapid response or reserve preservation
- flexibility or low cost
- production quality vs. availability of resources
- etc.

▷ Efficient Dispatch of Tertiary Reserve

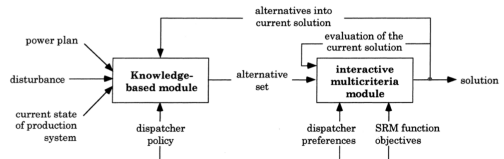
where the cooperation with the DM is as important as the optimisation engine itself

# Solution: CASTART, an interactive MCDSS combining AI and OR

- ▶ Sudden disturbance creates imbalance between production and consumption
- ▶ When automatic reserves are insufficient
- ▶ Dispatcher must restore system via tertiary reserve (SRM)



- ▶ Alternatives: possible actions on generation units
- ▶ Constraints: elapsed time, generation limits, ramp rates, reserve constraints,...
- ▶ Criteria: cost, contribution, destorage of reserve, reliability...



- ▶ A MCDSS to help in real-time the dispatcher in situations of disturbances
- ▶ Solution approach in two phases cooperating with the decision-maker
- ▶ MIP models and multi-objective heuristics, interactive multi-criteria selection procedure

# Some references

- ▶ X. Gandibleux, G. Libert, and P. Millot. On the multicriteria decision modelling of an on-line aid support in supervision of large industrial processes. In *Xth International Conference on Multiple Criteria Decision Making Proc.*, volume II, pages 241–250, 1992. July 19-24, 1992 Taipei, Taiwan, R.O.C.
- ▶ X. Gandibleux, C. Rosenthal-Sabroux, and G. Libert. Multicriteria approach for intelligent decision support in supervisory control. *Annals of Operations Research*, 51:327–347, 1994.
- ▶ X. Gandibleux. Interactive multicriteria procedure exploiting a knowledge-based module to select electricity production alternatives : the CASTART system. *European Journal of Operational Research*, 113(2):355–373, 1999.
- ▶ R. Allmendinger, M. Ehrgott, X. Gandibleux, M. Josef Geiger, K. Klamroth, and M. Luque. Navigation in multiobjective optimization methods. *Journal of Multi-Criteria Decision Analysis*, Volume 24, Issue 1-2, Pages 57-70, 2017.

MOCO, MOMH, MCDMLib  
(1995–)

---

# Multi-Objective Combinatorial Optimisation (MOCO)

$$\min_{S \in X} (z^1(S), \dots, z^p(S)) \quad (\text{MOCO})$$

- ▶  $A = \{a_1, \dots, a_n\}$
- ▶  $X \subseteq 2^A$
- ▶  $p \geq 2$

knapsack, assignment, covering, packing, partitionning, paths, flows, etc.

E. L. Ulungu. *Optimisation combinatoire multicritère: détermination de l'ensemble des solutions efficaces et méthodes interactives*. PhD thesis, Université de Mons-Hainaut, Belgium. 1993.

E. L. Ulungu and J. Teghem. Multi-objective combinatorial optimization problems: A survey. *Journal of Multi-Criteria Decision Analysis* 3:83–104. 1994

From 1996, with Matthias Ehrgott:

- ▶ Comprehensive review of the MOCO landscape
- ▶ Unified terminology and problem taxonomy
- ▶ Identified open problems that structured the research in the community
- ▶ Became the standard reference in the MCDM research community

M. Ehrgott and X. Gandibleux. A survey and annotated bibliography of multiobjective combinatorial optimization. *OR Spectrum*, 22:425–460, 2000.

# Multi-Objective Combinatorial Optimisation (MOCO)

$$\min_{S \in X} (z^1(S), \dots, z^p(S)) \quad (\text{MOCO})$$

- ▶  $A = \{a_1, \dots, a_n\}$
- ▶  $X \subseteq 2^A$
- ▶  $p \geq 2$

knapsack, assignment, covering, packing, partitionning, paths, flows, etc.

E. L. Ulungu. *Optimisation combinatoire multicritère: détermination de l'ensemble des solutions efficaces et méthodes interactives*. PhD thesis, Université de Mons-Hainaut, Belgium. 1993.

E. L. Ulungu and J. Teghem. Multi-objective combinatorial optimization problems: A survey. *Journal of Multi-Criteria Decision Analysis* 3:83–104. 1994

From 1996, with Matthias Ehrgott:

- ▶ Comprehensive review of the MOCO landscape
- ▶ Unified terminology and problem taxonomy
- ▶ Identified open problems that structured the research in the community
- ▶ Became the standard reference in the MCDM research community

M. Ehrgott and X. Gandibleux. A survey and annotated bibliography of multiobjective combinatorial optimization. *OR Spectrum*, 22:425–460, 2000.

# Some references (MOCO)

## ► **Book:**

M. Ehrgott and X. Gandibleux, editors. *Multiple Criteria Optimization: State of the Art Annotated Bibliographic Survey*, volume 52 of International Series in Operations Research and Management Science. 496 pages. Kluwer Academic Publishers, Boston, 2002.

## ► **Book chapter:**

M. Ehrgott and X. Gandibleux. "Multi-objective Combinatorial Optimisation: Concepts, Exact Algorithms and Metaheuristics". In *Fixed Point Theory, Variational Analysis, and Optimization* (S. A. R. Al-Mezel, F. R. M. Al-Solamy and Q. H. Ansari Eds). pp.307–341. Chapman and Hall/CRC, 2014.

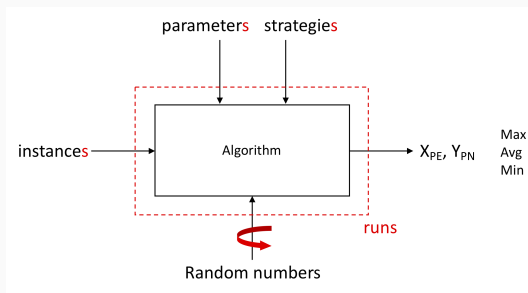
## ► **Special issue:**

M. Ehrgott, J. Figueira, and X. Gandibleux, guest editors. Special issue 'Multiple Objective Discrete and Combinatorial Optimization'. *Annals of Operations Research*, Volume 147. Number 1. Springer, October 2006.

# Multi-Objective Metaheuristic (MOMH)

## Definition

An **approximation solving methods** for a multi-objective optimisation problem **without performance guarantee a priori**, aiming to provide a good tradeoff between an approximation of the outcome set, and the time and memory requirements to obtain them.



M. Ehrgott and X. Gandibleux. Approximative Solution Methods for Multiobjective Combinatorial Optimization. *TOP: International Journal on Operations Research of the Spanish Society of Statistics and Operations Research*, Volume 12, Number 1, pp.1-63, June 2004. Springer Berlin.

# Multi-Objective Metaheuristic (MOMH)

## The pioneer's period by the MCDM community

inspired/extend IP/CO metaheuristics devoted to single objective optimization

1992: Acceptation rules for SA  
by Serafini.

1992: Multi-Objective Simulated Annealing (MOSA)  
by Ulungu and Teghem.

1996: Multiple Objective Tabu Search (MOTS)  
by Gandibleux, Mezdaoui, Fréville.

1997: Tabu Search for MOCO  
by Sun.

1998: GRASP for biSCP (MOGRASP)  
by Gandibleux, Vancoppenolle, Tuytens.

# Multi-Objective Metaheuristic (MOMH)

## The pioneer's period by the MCDM community

inspired/extend IP/CO metaheuristics devoted to single objective optimization

1992: Acceptation rules for SA  
by Serafini.

1992: Multi-Objective Simulated Annealing (MOSA)  
by Ulungu and Teghem.

1996: Multiple Objective Tabu Search (MOTS)  
by Gandibleux, Mezdaoui, Fréville.

1997: Tabu Search for MOCO  
by Sun.

1998: GRASP for biSCP (MOGRASP)  
by Gandibleux, Vancoppenolle, Tuytens.

GRASP, Greedy randomized adaptive search procedure (Fé0 and Resende, 1989):

$S^* \leftarrow \emptyset$ , the best solution found

**repeat**

$S \leftarrow \text{greedyRandomizedConstruction}(\text{problem}, \alpha)$

$S' \leftarrow \text{localSearchImprovement}(S)$

$\text{updateSolution}(S', S^*)$

**until**  $\text{isFinished?}(\text{StoppingRule})$

**return**  $S^*$

- ▶  $\alpha \in [0, 1]$ , the compromise between random and greedy.
- ▶  $\text{stoppingRule}$  (ex :  $\text{nIter}$ ,  $\text{elapsedTime}$ , etc.).

- Principle:

$$\circ d \left\{ \begin{array}{l} x_0 \leftarrow \text{construction}(\text{problem}, \alpha, \lambda^d) \\ x'_0 \leftarrow \text{localSearch}(x_0) \\ \circ i \left\{ \begin{array}{l} x \leftarrow \text{deconstruction}(x'_{i-1}) \\ x_i \leftarrow \text{reconstruction}(x, \lambda^d) \\ x'_i \leftarrow \text{localSearch}(x_i) \end{array} \right. \end{array} \right.$$

- Master thesis of D. Vancoppenolle

- MCDM'1998 Charlottesville (USA):

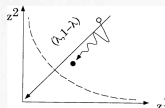
X. Gandibleux, D. Vancoppenolle, D. Tuytens, A first making use of GRASP for solving MOCO problems. MCDM'1998: 14th International Conference on Multiple Criteria Decision Making, June 8-12 1998, Charlottesville (VA), USA.

## MOGRASP

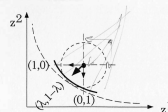
### ▷ GRASP construction

$$\lambda = (\lambda_1, \dots, \lambda_p) \geq 0$$

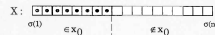
$$s_i(z, \lambda) = \sum_{j=1}^p \lambda_j z_j(x), \quad \sum_{j=1}^p \lambda_j = 1$$



### ▷ GRASP local search



ex : 1,0-exchange  
1,1-exchange  
1,2-exchange...



- Principle:

$$\circ d \left\{ \begin{array}{l} x_0 \leftarrow \text{construction}(\text{problem}, \alpha, \lambda^d) \\ x'_0 \leftarrow \text{localSearch}(x_0) \\ \circ i \left\{ \begin{array}{l} x \leftarrow \text{deconstruction}(x'_{i-1}) \\ x_i \leftarrow \text{reconstruction}(x, \lambda^d) \\ x'_i \leftarrow \text{localSearch}(x_i) \end{array} \right. \end{array} \right.$$

$\rightsquigarrow$  **LNS** (Shaw, 1998):

In LNS, an initial solution is gradually improved by alternately **destroying** and **repairing** the solution (Pisinger and Røpke 2010)

- Master thesis of D. Vancoppenolle

- MCDM'1998 Charlottesville (USA):

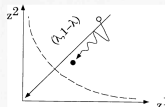
X. Gandibleux, D. Vancoppenolle, D. Tuytens, A first making use of GRASP for solving MOCO problems. MCDM'1998: 14th International Conference on Multiple Criteria Decision Making, June 8-12 1998, Charlottesville (VA), USA.

## MOGRASP

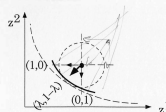
### ▷ GRASP construction

$$\lambda = (\lambda_1, \dots, \lambda_p) \geq 0$$

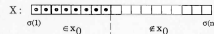
$$s_i(z, \lambda) = \sum_{j=1}^p \lambda_j z_j(x), \quad \sum_{j=1}^p \lambda_j = 1$$



### ▷ GRASP local search

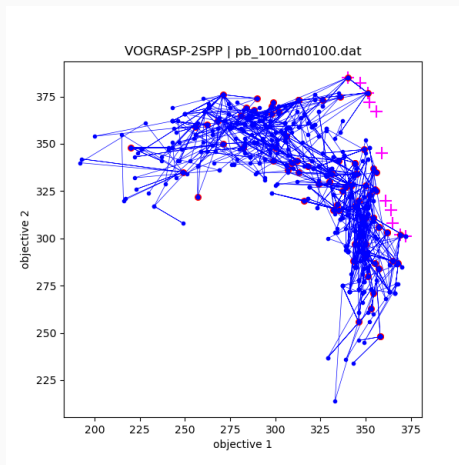


ex : 1,0-exchange  
1,1-exchange  
1,2-exchange...



## Illustration:

- ▶ Set Packing Problem:  
instance with 100 variables
- ▶ value for computing  $U_{limit}$  into RCL:  
 $\alpha = 0.7$
- ▶ # search directions from 0.0 to 1.0, step 0.01:  
 $nb\lambda = 100$
- ▶ # deconstruction/reconstruction:  
 $nbDR = 15$



# Multi-Objective Metaheuristic (MOMH)

## **The pioneers period** by the [EA community](#)

inspired/extend [Genetic Algorithms](#) devoted to single objective optimization

[1984: Vector Evaluated Genetic Algorithm \(VEGA\)](#)

by Schaffer.

[1989: The use of Pareto-ranking](#)

by Goldberg.

[1993: Multiple Objective Genetic Algorithm \(MOGA93\)](#)

by Fonseca and Fleming.

[1994: Nondominated Sorting Genetic Algorithm \(NSGA\)](#)

by Srinivas and Deb.

[1998: Strength Pareto Evolutionary Algorithm \(SPEA\)](#)

by Zitzler and Thiele.

# Multi-Objective Metaheuristic (MOMH)

## The hybridation's wave

1995: MOGA with Local search (MOGA-LS)

by Ishibuchi and Murata.

1996: Pareto Simulated Annealing (PSA)

by Czyzak and Jaskiewicz.

1997: Tabu Search for MOCO (TAMOCO)

by Hansen.

1998, 2001: GA, seeding solutions, repeated local search (MGK Algorithm)

by Morita, Gandibleux and Katoh.

2001: Multiple Objective Genetic Local Search (MO-GLS)

by Jaskiewicz.

2002: Multiple Objective Genetic Tabu Search (MOGTS)

by Barichard and Hao.

2003: Multi-Objective Path-Relinking

by Gandibleux, Morita, and Katoh.

# MOCO and MOMH: Some Applications

## Scheduling

Academic situation

### Bi-objective Flow-shop scheduling

J. Pasia, X. Gandibleux, K. Doerner and R. Hartl. Local Search Guided by Path Relinking and Heuristic Bounds. In Sh. Obayashi, K. Deb, C. Poloni, T. Hiroyasu and T. Murata Editors, *Evolutionary Multi-Criterion Optimization*, Volume 4403 of *Lecture Notes in Computer Science*, pp. 501-515, Springer, 2007.

## Railway transportation

Collaboration with [SNCF](#)

### Capacity of Railway Infrastructures

X. Gandibleux, P. Riteau, and X. Delorme. RECIFE: A MCDSS for Railway Capacity Evaluation. In *Multiple Criteria Decision Making for Sustainable Energy and Transportation Systems* (M. Ehrgott, B. Nadjoka, Th. Stewart, J. Wallenius Edts.), *Lecture Notes in Economics and Mathematical Systems* 634, pp.93-103, 2010.

## Logistics

Real-life situation ([Austrian Red Cross](#))

### Dial-a-ride problem for patient transport

S. N. Parragh, K. F. Doerner, R. F. Hartl, X. Gandibleux. A heuristic two-phase solution approach for the multi-objective dial-a-ride problem. *Networks*, Wiley, 54 (4), 227-242, 2009.

## Computer system

Real-life situation ([Google](#))

### Machine reassignment in data centres

T. Saber, X. Gandibleux, M. O'Neill, L. Murphy, A. Ventresque. A comparative study of multi-objective machine reassignment algorithms for data centres. *Journal of Heuristics*, 26, 119-150, 2020.

## Some references (MOMH)

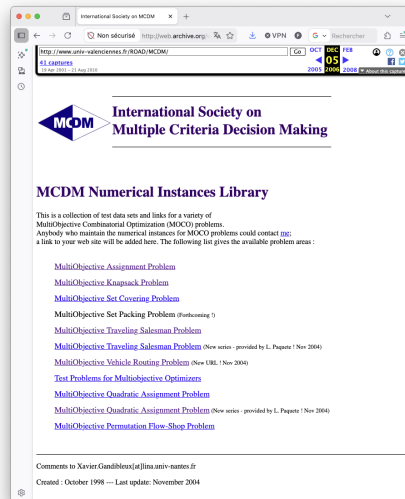
- ▶ X. Gandibleux, N. Mezdaoui, and A. Fréville. A tabu search procedure to solve multiobjective combinatorial optimization problems. In R. Caballero, F. Ruiz, and R. Steuer, editors, *Advances in Multiple Objective and Goal Programming*, volume 455 of *Lecture Notes in Economics and Mathematical Systems*, pages 291–300. Springer Verlag, Berlin, 1997.
- ▶ H. Morita, X. Gandibleux, and N. Katoh. Experimental feedback on biobjective permutation scheduling problems solved with a population heuristic. *Foundations of Computing and Decision Sciences Journal*, 26(1):23–50, 2001.
- ▶ X. Gandibleux, H. Morita, and N. Katoh. Evolutionary operators based on elite solutions for biobjective combinatorial optimization. In C. Coello Coello and G. Lamont, editors, *Applications of Multi-Objective Evolutionary Algorithms*, chapter 23, pages 555–579. World Scientific, Singapore, 2004.
- ▶ X. Gandibleux. Peek–Shape–Grab: a methodology in three stages for approximating the non-dominated points of multiobjective discrete combinatorial optimization problems with a multiobjective metaheuristic. In: H. Trautmann et al. (eds) *Evolutionary Multi-Criterion Optimization (EMO 2017)*, volume 10173 of *Lecture Note in Computer Science*, pages 221–235, Springer, 2017.

## MCDMLib (1998-2004) → MOCOLib (2007-2017)

- ▶ First public benchmark library for MOCO problems
- ▶ Instances available:  
assignment, knapsack, spanning tree,  
shortest path, TSP, ...
- ▶ Adopted as the reference in MCDM literature

## vOptLib (2017-)

- ▶ MOCO and MOMILP problems
- ▶ Github repository
- ▶ <https://voptsolver.github.io/vOptLib/>

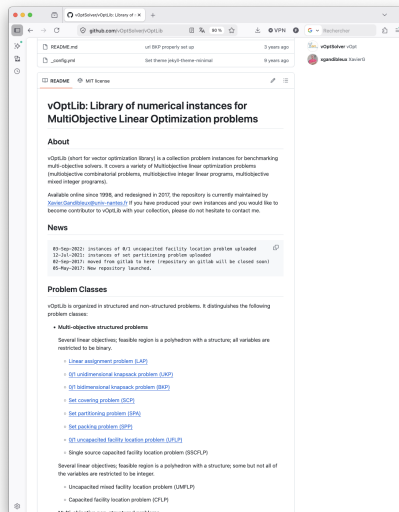


## MCDMLib (1998-2004) → MOCOLib (2007-2017)

- ▶ First public benchmark library for MOCO problems
- ▶ Instances available:  
assignment, knapsack, spanning tree, shortest path, TSP, ...
- ▶ Adopted as the reference in MCDM literature

## vOptLib (2017-)

- ▶ MOCO and MOMILP problems
- ▶ Github repository
- ▶ <https://voptsolver.github.io/vOptLib/>



$X_E$  and  $Y_N$ , Branch-and-Bound,  
Exact Algorithms (2000–)

---

# Studying $X_E$ and $Y_N$ (2000–)

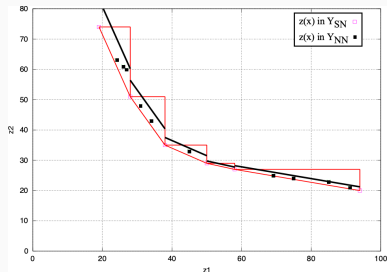
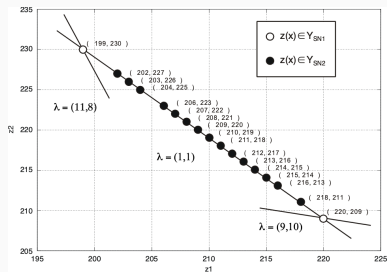
$$x \in X \subseteq \mathbb{R}^n \text{ and } y \in Y = z(X) \subseteq \mathbb{R}^p$$

## Motivations

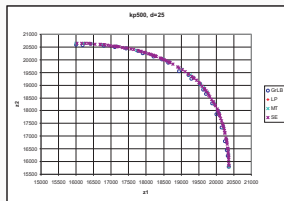
- Understanding (graal: characterizing)  $X_E$  and  $Y_N$  for designing efficient algorithms

## Some structural questions

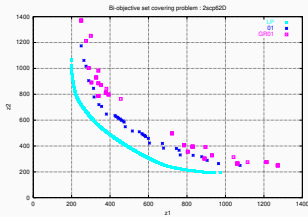
- How many efficient solutions exist?  
 $X_E, X_{E_m}, X_{E_M}$ ?
- How many nondominated points exist?  
 $Y_N, Y_{SN}, Y_{SN1}, Y_{SN2}, Y_{NN}$ ?
- How are they connected and distributed?  
Distribution of  $z(x)$  in  $Y$ ? Convex hull of  $Y_N$ ?  
(Dis)connected distribution of  $y \in Y_N$ ? Adjacency between nondominated points? Relaxations? Quality of relaxations? Etc.



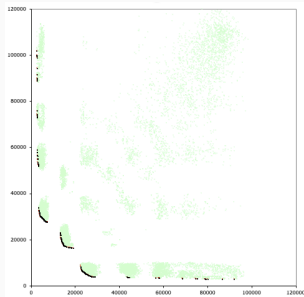
# Studying $X_E$ and $Y_N$ (2000–)



UKP: LP helpful; greedy helpful



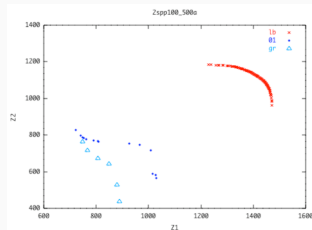
SCP: LP useless (but not always)



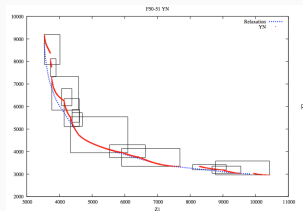
SPA:  $Y_{PN}$  not connected; gaps in  $Y$

## MCDM'2002 Semmering (AT):

X. Gandibleux, M. Ehrgott. Bounds and Bound Sets for Biobjective Combinatorial Optimization Problems: set covering, set packing and set partitioning problems with two objectives. 16th International Conference on Multiple Criteria Decision Making, Semmering, Austria, February 18-22, 2002.



SPP: LP convex, useless;  $Y_N$  not convex, not connected;  $\#Y_{SN} \approx \#Y_{NN}$



UFLP: many local  $Y_{SN}$

## What is a bound in a multi-objective Branch-and-Bound?

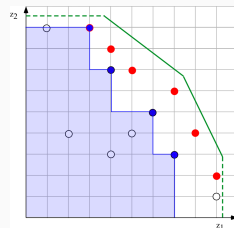
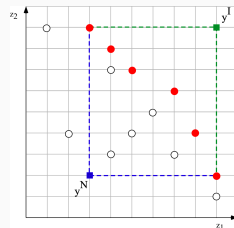
- ▶ In SO-B&B: a lower ( $\bar{v}$ ) and upper ( $\underline{v}$ ) value around  $f(x^*)$
- ▶ In MO-B&B: a lower ( $y^I$ ) and upper ( $y^N$ ) bound around  $Y_N$

MCDM'2000 Ankara (TR):

M. Ehrgott and X. Gandibleux. Bounds and bound sets for biobjective combinatorial optimization problems. In M. Koksalan and S. Zionts, editors, Multiple Criteria Decision Making in the New Millennium, volume 507 of Lecture Notes in Economics and Mathematical Systems, pages 241–253. Springer, 2001.

## MO-B&B: from scalar bounds to **bound sets**

- ▶ a **lower bound set (LBS)**  
e.g. relaxation giving a lower approximation of  $Y_N$
- ▶ an **upper bound set (UBS)**  
e.g. a current set  $S_N$  giving an upper approximation of  $Y_N$



## Some references (MO-B&B)

- ▶ M. Ehrgott and X. Gandibleux. Bound Sets for Biobjective Combinatorial Optimization Problems. *Computers and Operations Research*. Volume 34, Issue 9, pages 2674-2694. September 2007.
- ▶ A. Cerqueus, A. Przybylski, and X. Gandibleux. Surrogate upper bound sets for bi-objective bi-dimensional binary knapsack problems. *European Journal of Operational Research*, 244(2):417–433, July 2015.
- ▶ A. Przybylski, X. Gandibleux. Multi-Objective Branch and Bound. *European Journal of Operational Research*, Volume 260, Issue 3, Pages 856-872, 2017.
- ▶ X. Gandibleux, G. Gasnier, S. Hanafi. A primal heuristic to compute an upper bound set for multi-objective 0-1 linear optimisation problems. In: Proceedings of MODeM2021: Multi-Objective Decision Making Workshop, Hayes, Mannion, Vamplew (eds.) July 14-16, 2021. Online conference (<http://modem2021.cs.nuigalway.ie>). 8 Pages, available at: [http://www.optimization-online.org/DB\\_HTML/2021/07/8508.html](http://www.optimization-online.org/DB_HTML/2021/07/8508.html)

## Problem-Specific algorithms

- Philosophy: exploit problem structure

Example: **Two-phase method**

Phase 1: compute  $Y_{SN}$

Phase 2: compute  $Y_{NN}$

**Problems addressed:**

2-LAP, 3-LAP, 2-UKP, 3-UKP, 2-01UFLP,  
2-MUFLP, 2-TSP-D,  $(p - \sum | 1\text{-MinMax})$   
shortest path, etc.

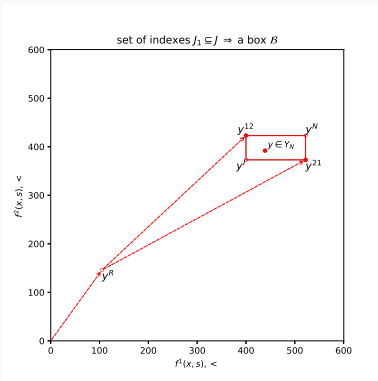
## Problem-Generic algorithms

- Philosophy: address any instance belonging in a given class of optimization problems

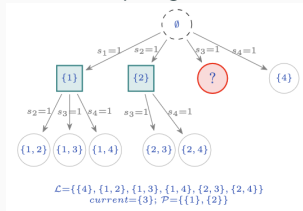
**Class of problems addressed:**

**MO-B&B** algorithm to compute  $Y_N$  for MOMILP. **Weight-set decomposition** algorithm to compute  $Y_{SN1}$  for MOIP.

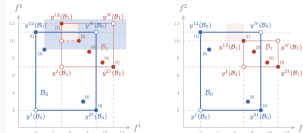
# Paving and computing $Y_N$ for 2-01UFLP



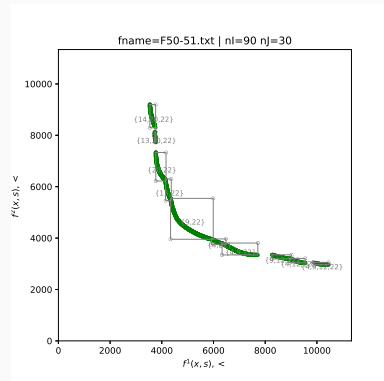
## Phase 1: the paving



## Phase 2: the refinement



## Phase 3: the generation



## Some references (Exact Algorithms)

- ▶ X. Gandibleux, F. Beugnies and S. Randriamasy. Martins' algorithm revisited for multi-objective shortest path problems with a MaxMin cost function. *4OR: Quarterly journal of Operations Research*. 4(1):47–59, March 2006.
- ▶ A. Przybylski, X. Gandibleux, and M. Ehrgott. A recursive algorithm for finding all extremal supported nondominated points in the outcome set of a multi-objective integer linear problem. *INFORMS Journal on Computing*, 22 (3), 371-386. 2010.
- ▶ T. Vincent, F. Seipp, S. Ruzika, A. Przybylski, and X. Gandibleux. Multiple objective branch and bound for mixed 0-1 linear programming: Corrections and improvements for the biobjective case. *Computers & Operations Research*, 40(1):498–509, 2013.
- ▶ X. Gandibleux, and A. Przybylski. Paving and computing the set of nondominated points for the bi-objective 0/1 uncapacitated facility location problem. Optimisation-online (<https://optimization-online.org/?p=34431>). Published: 2026/04/18.

## IP Routing

Collaboration with Alcatel

### Multi-objective QoS routing in IP networks

S. Randriamasy, X. Gandibleux, J. Figueira, and P. Thomin. *Device and a method for determining routing paths in a communication network in the presence of selection attributes*. Patent 11/25/2004. #20040233850. Washington, DC, USA..

## Energy policy

Real-life situation

### Electric Vehicle Charging

E. De Schepper, S. Van Passel, S. Lizin, T. Vincent, B. Martin, and X. Gandibleux. Economic and environmental multi-objective optimisation to evaluate the impact of Belgian policy on solar power and electric vehicles. *Journal of Environmental Economics and Policy*, 5(1):1–27, 2016.

## Open-Source Software (2014–)

---

## 2003: My Habilitation thesis

2.3. Conclusions et perspectives (#52)

Xavier Gandibleux

### PERSPECTIVES

⇒ **Elaboration de connaissances en vue de la construction de procédures de calcul exactes**

- Etude de propriétés caractéristiques des MOCO (seront davantage vers la frontière efficace, prétraitements, bornes, etc.)
- A terme : traitements spécifiques pour les problèmes de grandes tailles (quel peu abordé lors contexte des applications réelles)
- Projet : élaboration d'une librairie de codes, à l'image d'un 'VO-cplex'
- Thèse : A. Puyhydt, F. Desgripes
- Collaborations : M. Ellegott, R. Wolpert
- Papiers en préparation : bornes (COR) ; affectations; routage IP (ACOR)



12 décembre 2003

### Perspectives

⇒ Developing knowledges in exact computation procedures

### Project:

Development of a code library, such as a 'VO-cplex'

2015: Kick-off of the DFG/ANR research project vOpt

2017: Introduction of [vOptSolver](#) (vOptGeneric, vOptSpecific)

X. Gandibleux, G. Soleilhac, A. Przybylski, F. Lucas, S. Ruzika, P. Halffmann. vOptSolver, a "get and run" solver of multiobjective linear optimization problems built on Julia and JuMP. *MCDM2017: 24th International Conference on Multiple Criteria Decision Making*. July 10-14, 2017, Ottawa, Canada.

2019: First discussions on discourse/new design of MOO in JuMP

2023: Introduction of [MultiObjectiveAlgorithms.jl](#)

- part concerning the MO model → [JuMP.jl](#)

- part concerning MO algorithms → [MultiObjectiveAlgorithms.jl](#)

G. Kof, O. Dowson, X. Gandibleux. From vOptGeneric.jl to MultiObjectiveAlgorithms.jl  
JuliaDays-Paris'2023: Int. workshop "Julia and Optimization", 4-6 October, 2023, Paris, France.

O. Dowson, X. Gandibleux and G. Kof. MultiObjectiveAlgorithms.jl: a Julia package for solving multi-objective optimization problems. *INFORMS Journal on Computing*, 2026. Accepted 04/04/2026, Available soon for download at <https://github.com/INFORMSJoC/2025.1449>

2015: Kick-off of the DFG/ANR research project vOpt

2017: Introduction of [vOptSolver](#) (vOptGeneric, vOptSpecific)

X. Gandibleux, G. Soleilhac, A. Przybylski, F. Lucas, S. Ruzika, P. Halffmann. vOptSolver, a "get and run" solver of multiobjective linear optimization problems built on Julia and JuMP. *MCDM2017: 24th International Conference on Multiple Criteria Decision Making*. July 10-14, 2017, Ottawa, Canada.

2019: First discussions on discourse/new design of MOO in JuMP

2023: Introduction of [MultiObjectiveAlgorithms.jl](#)

- part concerning the MO model → [JuMP.jl](#)
- part concerning MO algorithms → [MultiObjectiveAlgorithms.jl](#)

G. Kof, O. Dowson, X. Gandibleux. From vOptGeneric.jl to MultiObjectiveAlgorithms.jl  
JuliaDays-Paris'2023: Int. workshop "Julia and Optimization", 4-6 October, 2023, Paris, France.

O. Dowson, X. Gandibleux and G. Kof. MultiObjectiveAlgorithms.jl: a Julia package for solving multi-objective optimization problems. *INFORMS Journal on Computing*, 2026. Accepted 04/04/2026, Available soon for download at <https://github.com/INFORMSJoC/2025.1449>

# JuMP and MultiObjectiveAlgorithms (MOA)

## Open-source multi-objective software

- modeling ↔ solving ↔ analyzing

Any user can solve MOMP with JuMP  
and just a few lines of Julia code.

## Algorithms

- Hierarchic, Lexicographic, Dichotomic,  
 $\varepsilon$ -constraint, Chalmet, KirlikSayin,  
DominguezRios, TambyVanderpooten ...

## Solvers

- open-source: GLPK, HiGHS, ...  
commercial: Gurobi, CPLEX, ...

```
using JuMP, Gurobi
import MultiObjectiveAlgorithms as MOA

function biUKP(C::Matrix, w::Vector, b::Int, algorithm)
    model = Model(() -> MOA.Optimizer(Gurobi.Optimizer))
    set_attribute(model, MOA.Algorithm(), algorithm)
    @variable(model, x[1:length(w)], Bin)
    @objective(model, Max, C * x)
    @constraint(model, w' * x <= b)
    optimize!(model)
    return [ ( value.(x; result = i),
               objective_value(model; result = i)
             ) for i in 1:result_count(model) ]
end
```

---

```
julia> C, w, b = rand(1:20, 2, 5), rand(1:20, 5), 25
julia> biUKP(C, w, b, MOA.EpsilonConstraint())
```

---

```
([0.0, 0.0, 1.0, 1.0, 1.0], [33.0, 22.0])
([0.0, 1.0, 1.0, 1.0, 0.0], [32.0, 38.0])
([1.0, 1.0, 1.0, 0.0, 0.0], [15.0, 45.0])
```

## Now and Tomorrow (2026–)

---

## Generic & Specific Algorithms

Branch-&-xx, p-xFLP & beyond, p-TSP-d, ...

## (Meta)heuristics

Primal heuristics, matheuristics combining exact and heuristic methods, ...

## Software & Applications

MOLP, MOMILP, vOptSpecific, ...

# Many thanks

Local collaborator

- ▶ Anthony Przybylski

MO-Remote collaborators...

- ▶ Matthias Ehrgott, Sophie Parragh, Antonio Nebro, Oscar Dowson, Gökhan Kof, Saïd Hanafi, Stephan Ruzika, Pascal Halffmann, Andrzej Jaszkievicz, Vincent T'Kindt, Arnaud Fréville, Daniel Tuytens, Sabine Randriamasy, Hiroyuki Morita, Naoki Kato, . . . .

...MO-Friends

- ▶ RAMOO/MCDM/EWG MCDA/MOPGP (Kathrin, Thomas, Lars, Sune, Nicolas, Luis, Ralph, Daniel, Alexis, Roman, Yanis, Kaisa, Murat, Serpil, Margaret, Salvatore, José, Philippe, Francisco, Dylan, Fouad,...), MIC (Eric, Marc, Karl, Ibrahim, Fred, Celso,...), . . . .

PhD and Master students

- ▶ Xavier, Anthony, Fabien, Julien, Daniel, Hugo, Aurélien, Benoit, Thomas, Ellen, Audrey, Quentin; Nazik, David, Frédéric, Nadarajen, Takfarinas, Olga, Benjamin, Salim, Alban, Axel, Gauthier, Owein, Elizabeth, Awen, Valentin and many others.

. . . and all those who are not mentioned due to limited space!

# A Picture and a Wish to Conclude



S. Zionts,  
Y. Haimes,  
R. Steuer



Taken in 1992 @ the *10th Int. Conf. on Multiple Criteria Decision Making*, Taipei, Taiwan, R.O.C.:

Stanley Zionts  
Milan Zeleny and Po-Lung Yu  
Andrzej P. Wierzbicki

MCDM Gold Medal  
Edgeworth-Pareto Award  
Georg Cantor Award